2 Social Constructivism

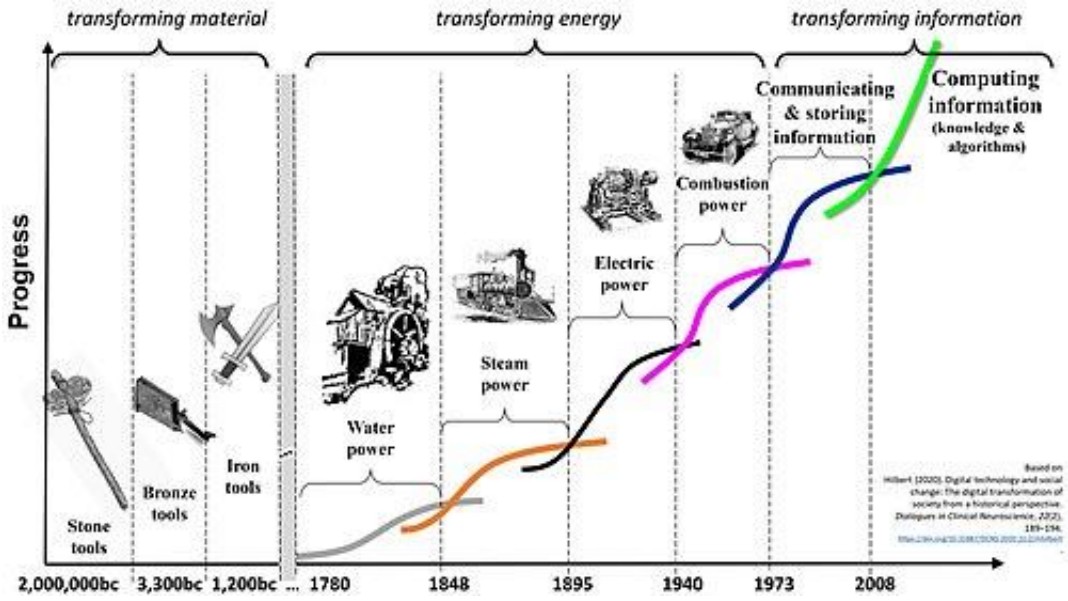
GEB2508 AI and Society

Content

- Historiography of Science
  - Internalism vs. Externalism
  - The Sociological Turn
- Social Construction of Facts
  - SSK and The Strong Programme
  - Social Construction Theory
- Social Construction of Artifacts
  - Technological Determinism
  - Social Construction of Technology

Progressive Narrative

- The development of S&T is presented as a tale of **linear, upward, cumulative** progress.
  - Technological optimism and determinism
- Critiques of the progressive narrative
  - Kuhn's paradigm shifts
  - Social constructivism
  - Global and decolonial histories



Internalism vs. Externalism

Internalism

- The history of scientific ideas, experiments, and discoveries is driven by the **internal** logic and intellectual needs of science itself
- Development of science as an autonomous, self-contained process

Externalism

- Science is a socially embedded activity, and scientific developments are **contingent** on external factors.
- Examining how external factors shape scientific knowledge
- **Interaction** between science and its broader context

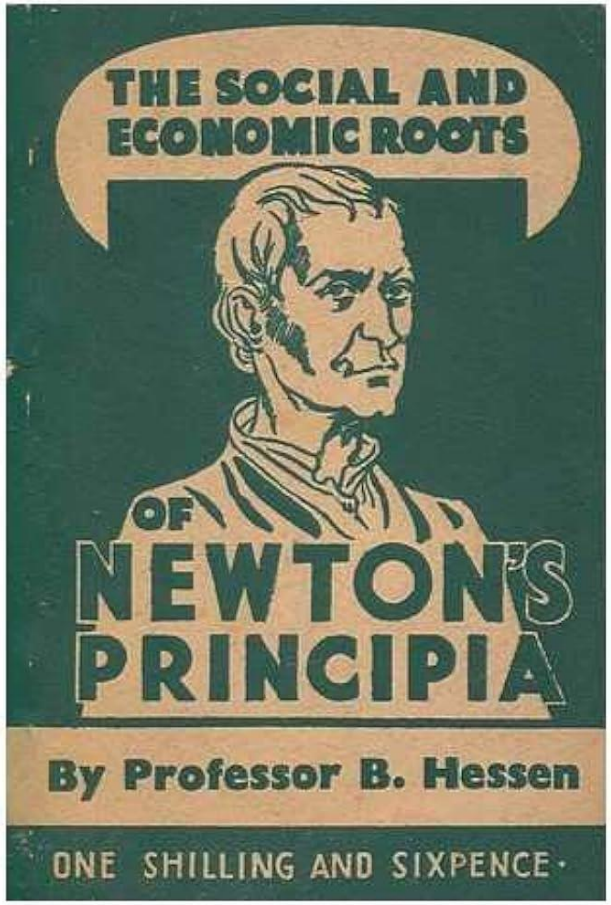
The Case of Darwin’s Theory of Evolution

Internalist Explanation

Externalist Explanation

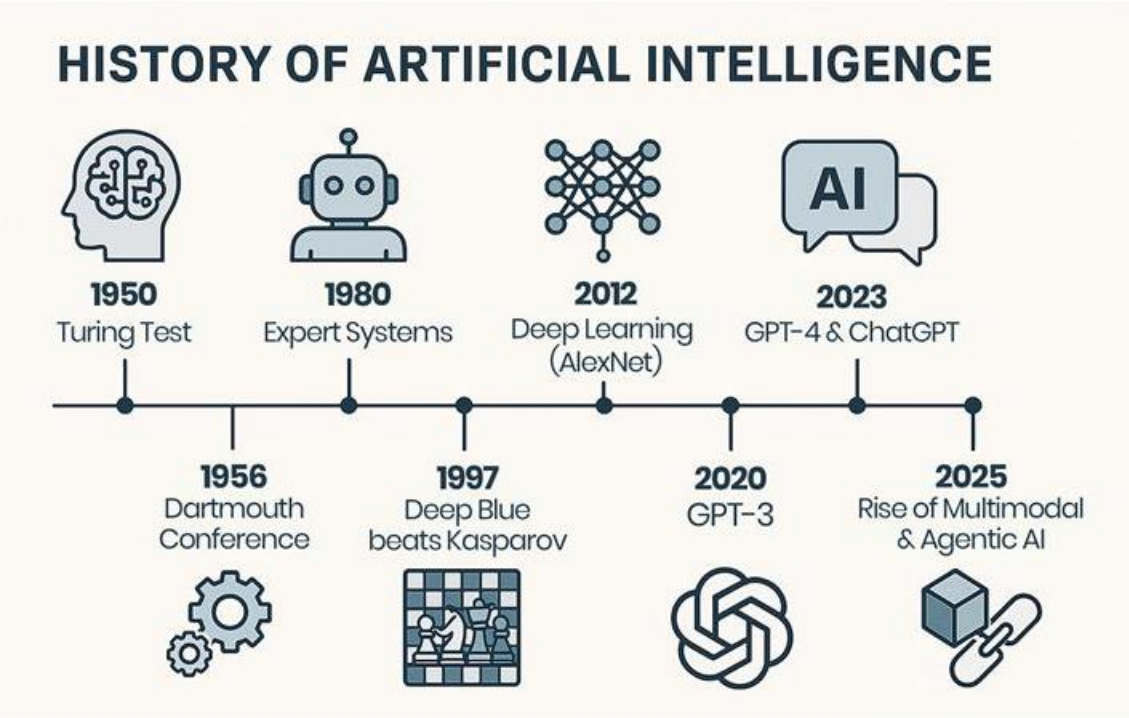
Newton's *Principia* From An Externalist View

1. Relationship between science and the **socio-economic development** of early capitalism through Newton's *Principia*
2. Limitations of 17th-century scientists: they could not make advancements in fields like thermodynamics or electromagnetism.
3. Impact of **ideological** constraints during the Glorious Revolution: Newton had to reconcile his materialistic perspective with divine intervention.



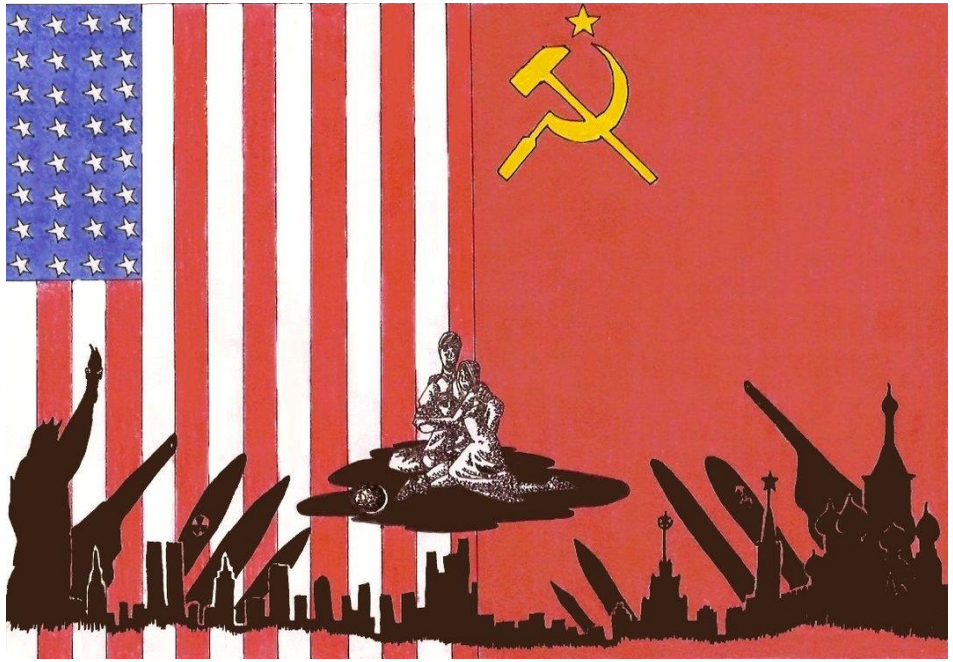
AI History From An Internalist View

- Birth of AI as a field: Turing test (1950), Dartmouth Conference (1956), Perceptron (1957),
- Machine learning & deep learning: Deep Blue (1997), Deep Learning (2010s)
- AI development as results of algorithms, data structures, and computing power



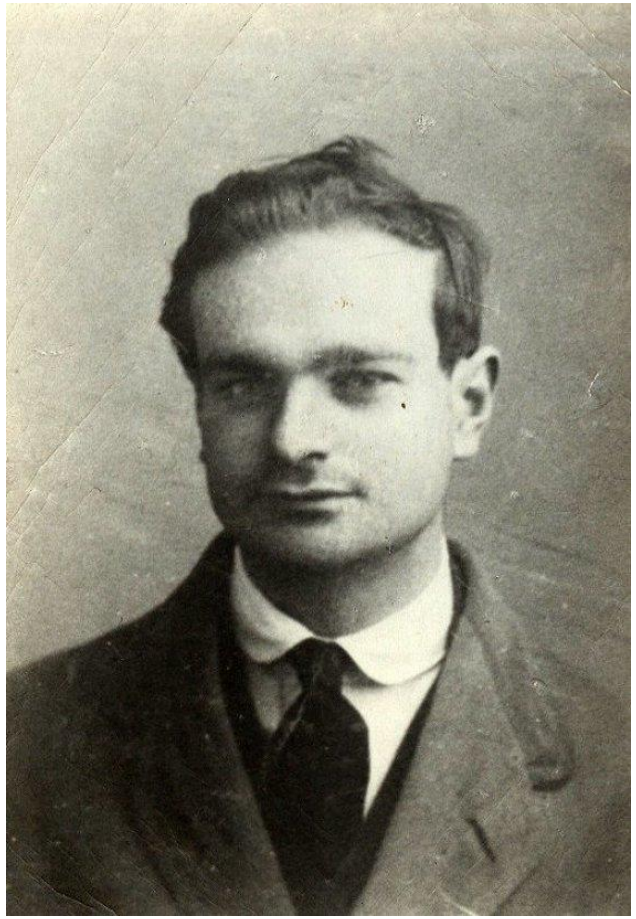
AI History From An Externalist View

- Cold War origins and military funding
  - The Defense Advanced Research Projects Agency (DARPA) (1958)
- Revival driven by tech industry and global economy
  - U.S.-Japan rivalry over the Fifth Generation Computer Systems (FGCS) in the 1980s
- Contemporary AI boom and geopolitics
  - U.S.-China rivalry



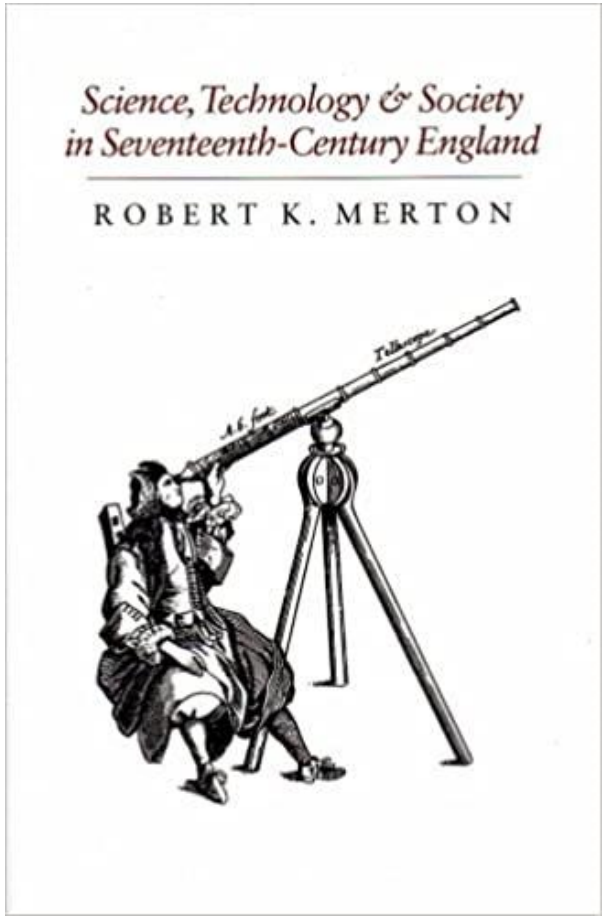
Boris Hessen (1893–1936)

- Hessen's background
  - Role in soviet science; political persecution
  - Taught at the Institute of History of Science and Technology of Moscow University
- **Marxist** approach to the history of science
  - Hessen's 1931 paper
  - Cornerstone of the **externalist** approach
- Hessen's legacy and impact
  - Joseph Needham and Robert Merton
  - Edinburgh School and SSK



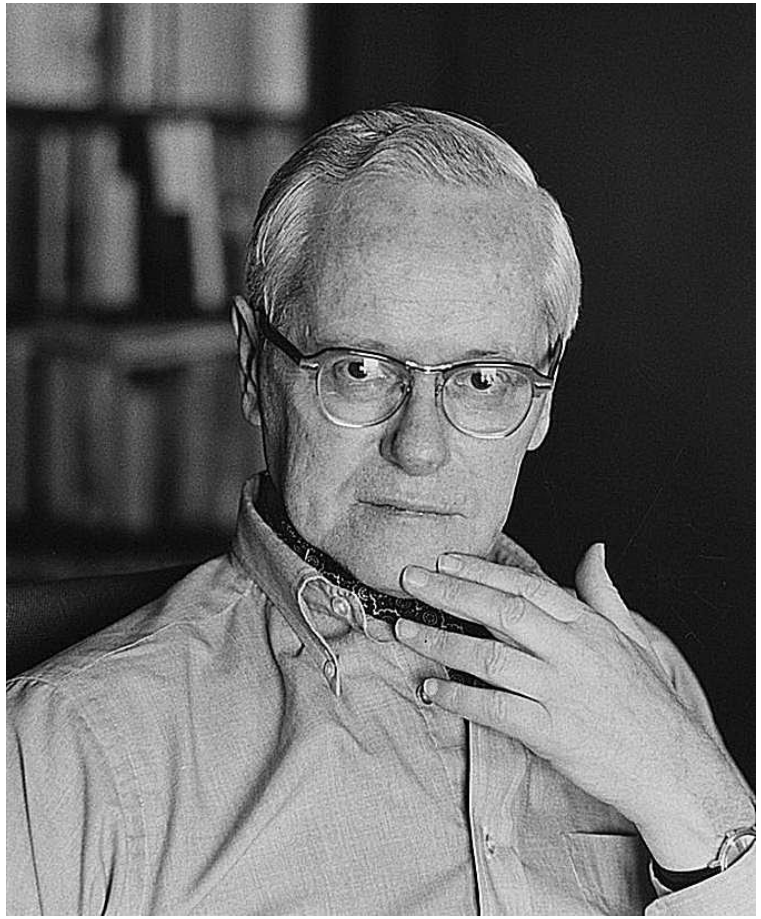
Cultural and Religious Influences

- Robert Merton (1938) *Science, Technology and Society in 17th Century England*
- Role of religion
  - Puritan ethos and science
  - **Justification** of science through religion
  - Compatibility with scientific practices
- Methodological contribution
  - **Institutional** analysis (e.g., Royal Society)
  - Sociology of science



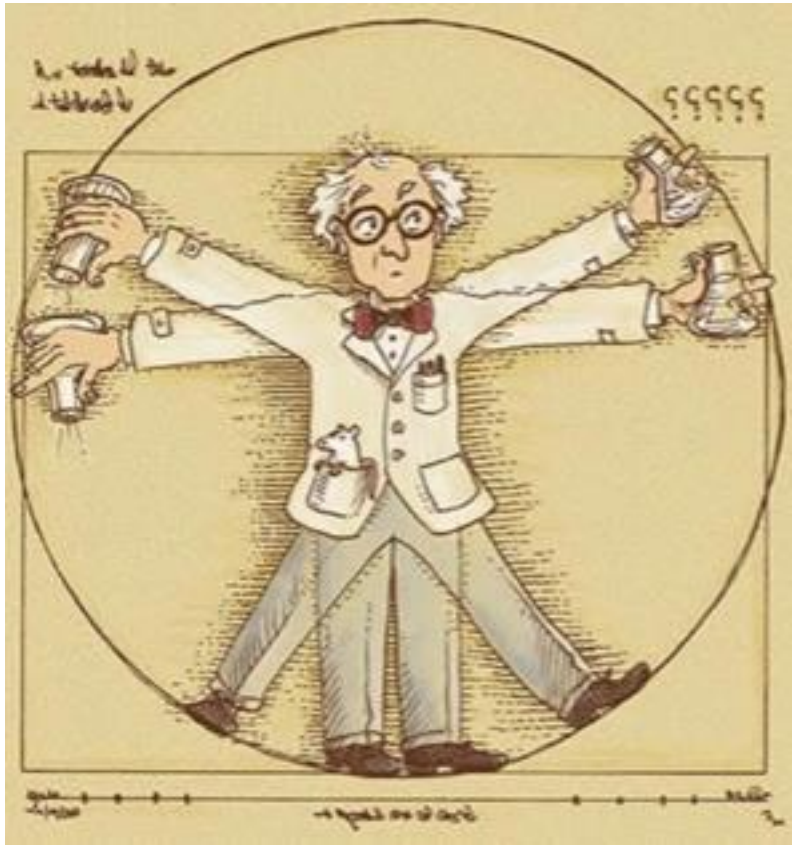
Robert Merton (1910-2003)

- Founder of the Sociology of Science
- PhD in sociology at Harvard
  - Studied under Talcott Parsons
- **Normative** structure of science (CUDOS)
  - Communism
  - Universalism
  - Disinterestedness
  - Organized Skepticism
- Institutional role of science



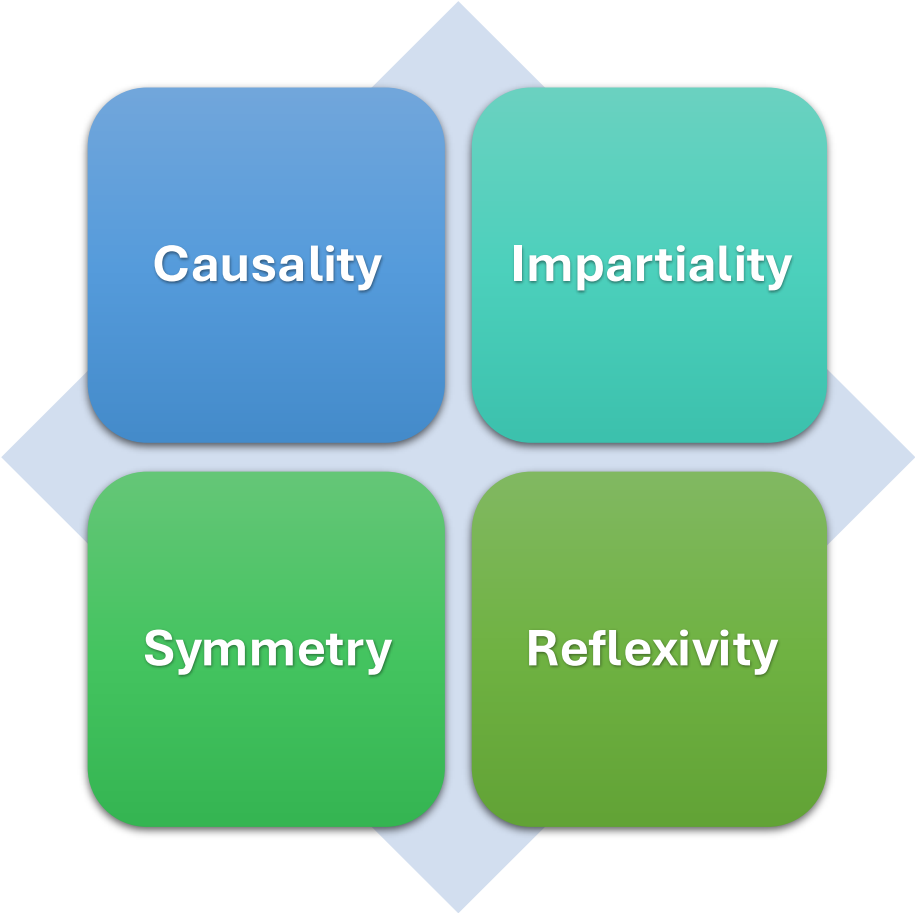
The Sociological Turn

- Departure from internalist perspectives during the mid-20th century
  - Knowledge as socially constructed
  - Emergence of the field of STS
- Key figures
  - Boris Hessen (1893-1936); Robert Merton (1910-2003); Thomas Kuhn (1922-1996)
  - Bruno Latour (1947-2022) and Actor-Network Theory (ANT)
  - David Bloor (1942-) and the Sociology of Scientific Knowledge (SSK)



From 'Weak' to 'Strong'

- The Strong Programme was formulated by David Bloor in *Knowledge and Social Imagery* (1976)
- The sociology of scientists vs. the sociology of the **content** of science.
- Sociology of Scientific Knowledge (SSK) goes beyond studying social influences and investigates the **inherent social construction** of scientific knowledge.



The Edinburgh School (1970s-80s)

- The Strong Program originated in the Science Studies Unit at the University of Edinburgh, established in 1964 as an initiative to **broaden** undergraduate science education.
- Key figures include the sociologist Barry Barnes, the philosopher David Bloor, and the historian Steven Shapin

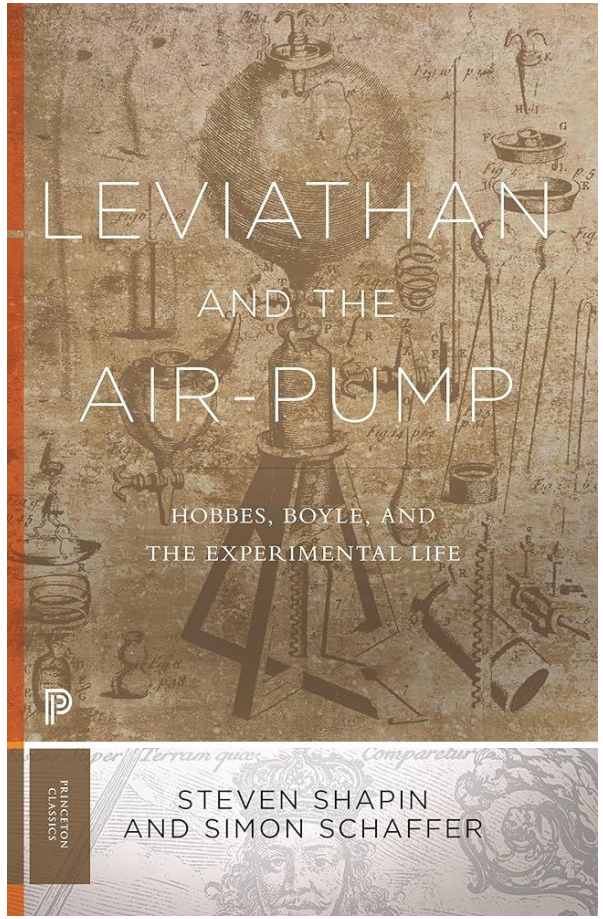
The Case of the Boyle-Hobbes Debate

- Robert Boyle (1627–1691) championed **empiricism**, arguing that knowledge should be based on observations and experiments, and experiments should be **witnessed** to establish **credibility**.
- Thomas Hobbes (1588–1679) argued that experiments were prone to error and **subjective** interpretation, and believed knowledge should be based on **deductive reasoning**.

An Experiment On A Bird In The Air Pump by Joseph Wright of Derby

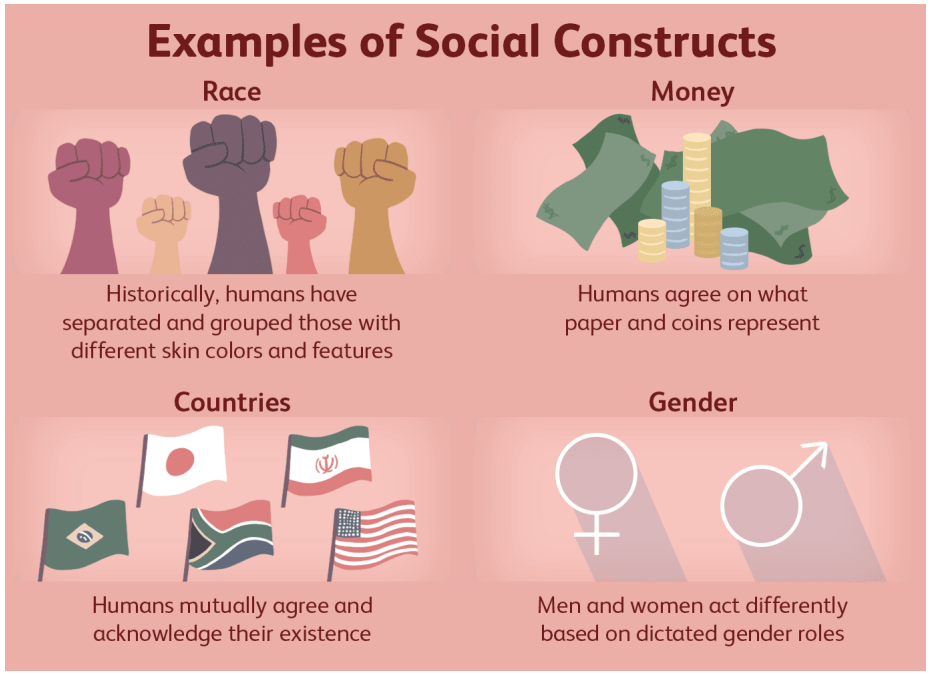
Leviathan and the Air-Pump (1985)

- Theme: how **experimental** science became an accepted and authoritative way of understanding nature
- Science as a social process: social practices surrounding experimental science
- Science and democracy: public witnessing for **transparency** and **accountability**
  - Hobbes' critique on **decentralized** knowledge and **elitist** experimental science



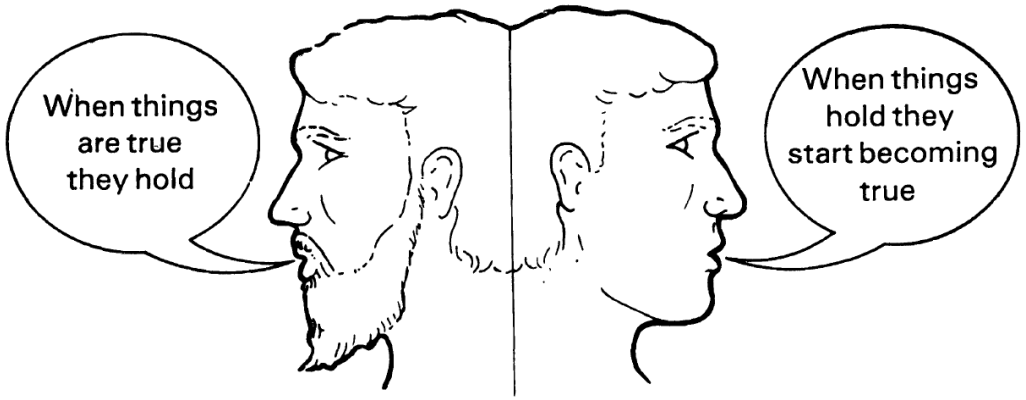
Social Construction Theory

- Essentialism (something is **natural**) vs. social constructivism (a taken-for-granted **invention**)
- Reality is socially constructed through collective processes, influenced by power dynamics and authority.
- Language and symbols play a central role in shaping these constructions.
- Inevitability vs. **contingency**



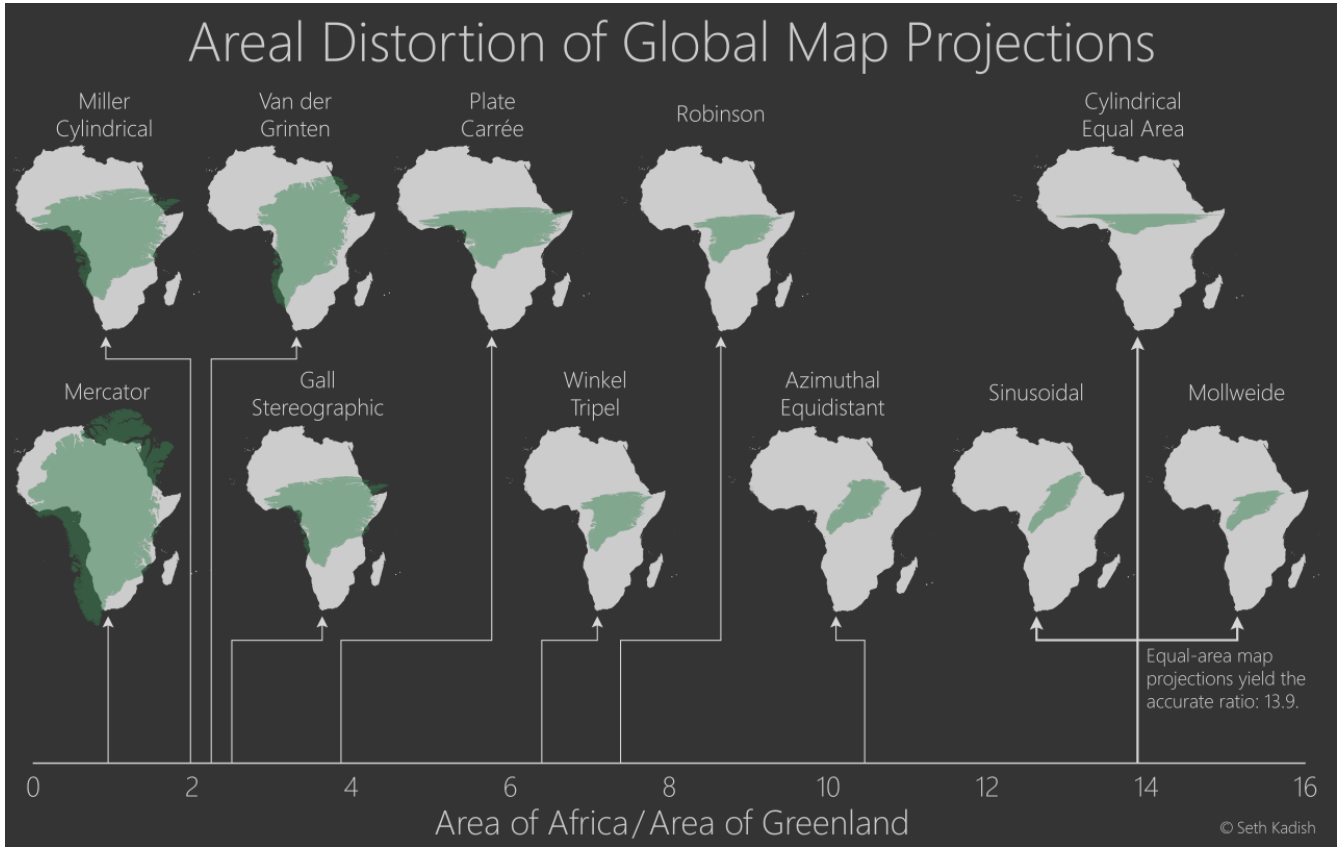
Social Construction of Scientific Facts

- The production of facts involves social **negotiation**
  - E.g., the theory of phlogiston
- Facts as outcomes of debates rather than discoveries
  - E.g., the classification of Pluto
- Knowledge as **contingent** on specific contexts
  - E.g., the concept of mental illness



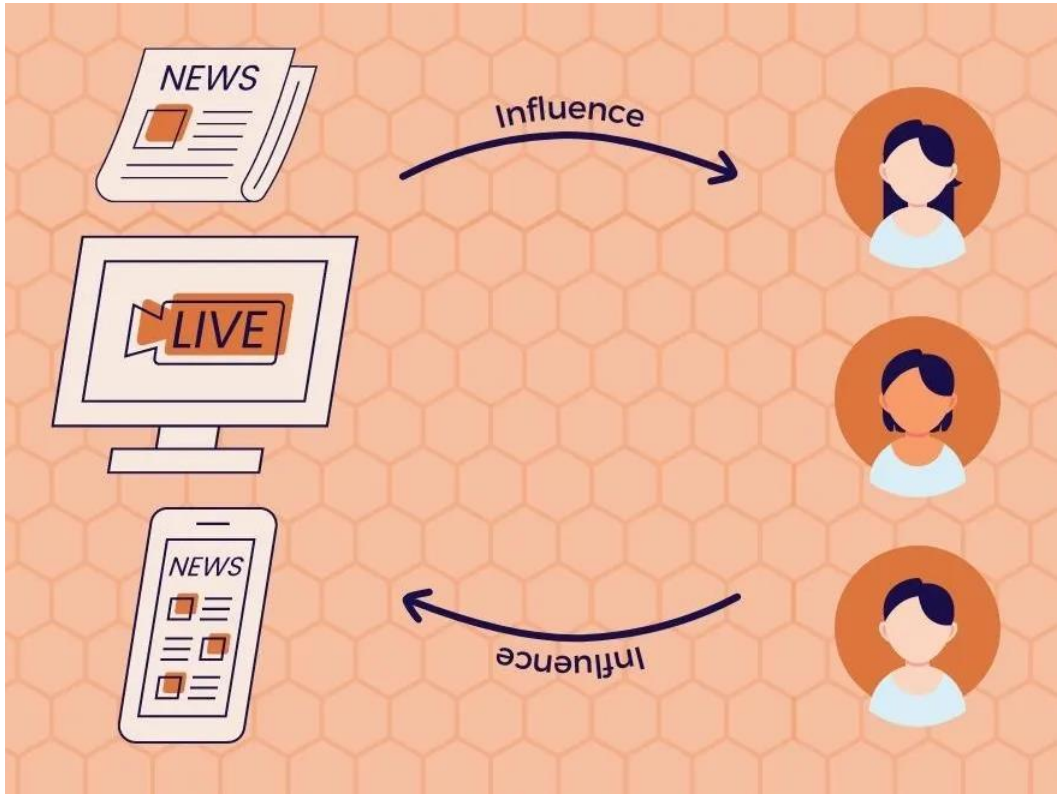
The Case of Map Projections

- In what way do you think the world map is socially constructed?
- What types of values and whose values have shaped the selection of map projections?



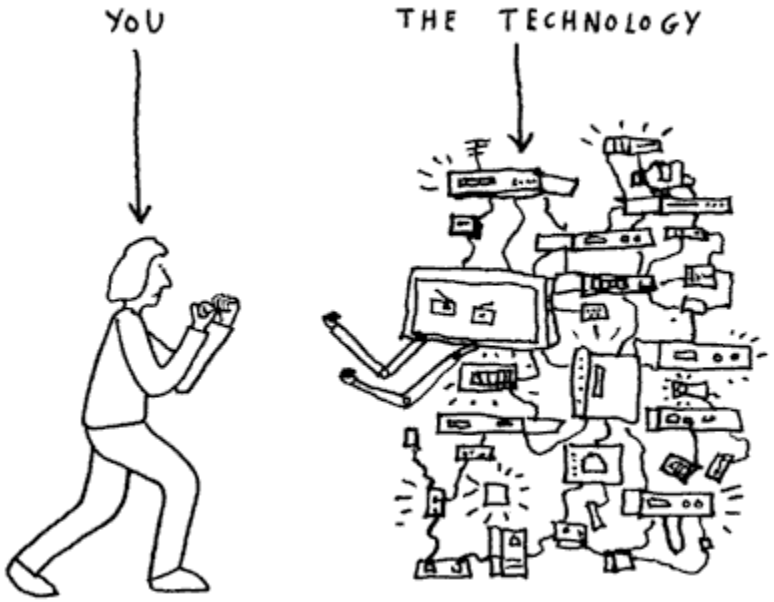
Relationship Between Technology & Society

- Technological Determinism
  - Technology → Society
  - Technology is the primary driver of societal change.
- Technological Constructivism
  - Technology ← Society
  - Technology is a product of social negotiation and not an autonomous force.



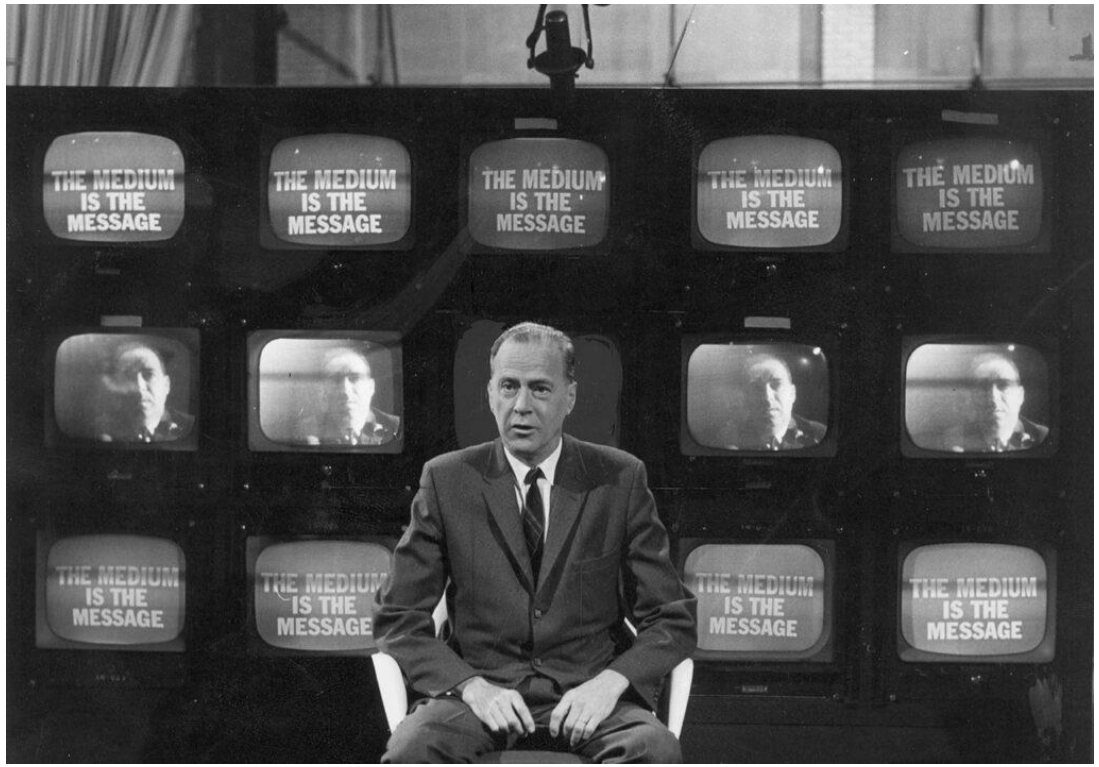
Technological Determinism

- Technology is regarded as an **autonomous** force that drives social change.
- Thorstein Veblen describes technology as the **prime mover** in history.
- Marx sees technology as the key driver of changes in the modes of production.
  - “The windmill gives you society with the feudal lord: the steam mill with the industrial capitalist.” (Karl Marx, The Poverty of Philosophy)



The Case of Media

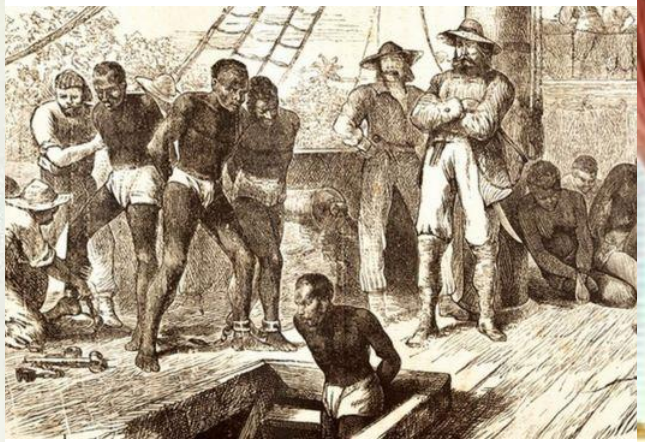
- “The medium is the message.”
- The **form** of media is more important than the **content**.
  - The printing press leads to individualism and nationalism
  - Electronic media create the global village, collapsing space and time.
- Media as the environment taken for granted

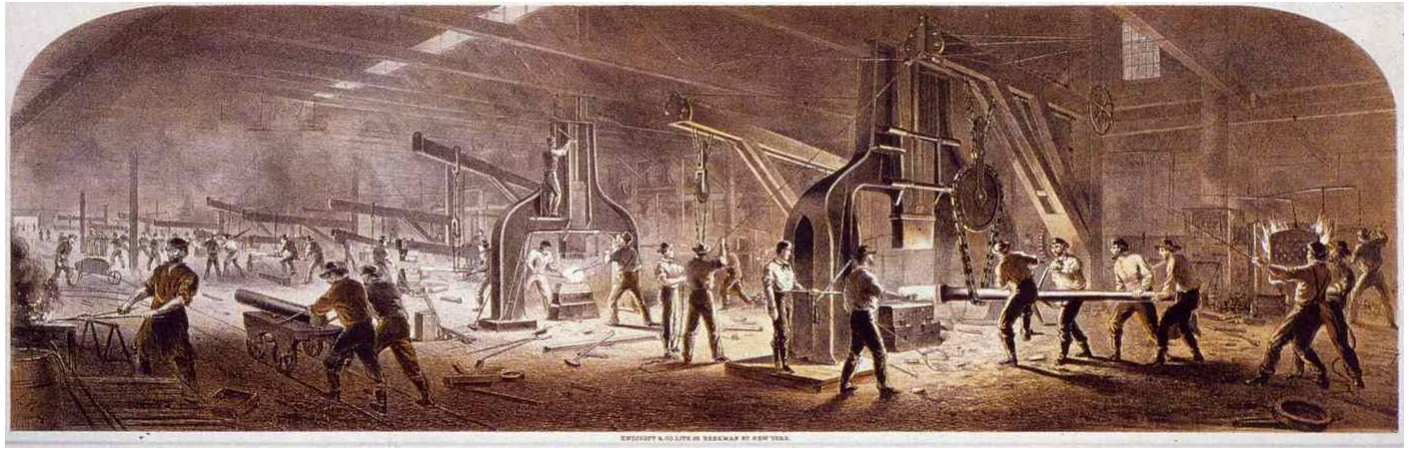


Marshall McLuhan (1911-1980)

The Case of Abolition

- Technology (the steam engine) drives economic change
- Economic transformation shapes morality and politics

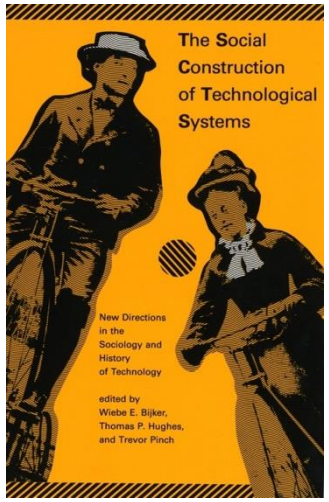




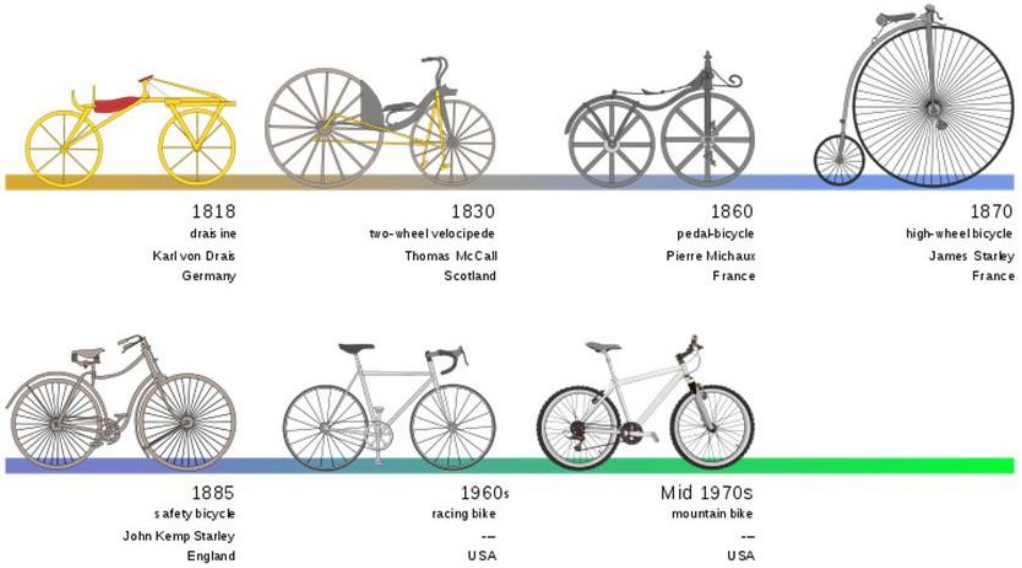
Social Construction of Technology (SCOT)

- Technological constructivism: technology as a social product
- **Interpretive flexibility**: how relevant social groups ascribe **different meanings** to an artifact → contest and negotiation
- Technologies do not succeed based on some **objective** measure of their inherent quality or efficiency; rather, they are adopted because they are perceived to align with and advance the **interests** of certain stakeholders.

The Case of The Bicycle



- Development and adoption of the bicycle as a classic example of technological constructivism
- Early designs and social negotiations: from the high-wheeler to the safety bicycle
- The design was shaped by the demands and feedback of **various** social groups.



Cultural Symbol of Women’s Liberation

- Controversies: Should a bicycle be considered masculine or feminine? Is it appropriate for women to ride bicycles?
- An instrument of **empowerment**: bicycles symbolized women’s freedom, independence, and mobility.
- Impact on women’s **fashion**: part of a broader movement toward more practical and less restrictive clothing

The Case of AI-Generated Hands

- What insights does the “AI hands” problem provide into the social constructivist understanding of technology?